Experimental Methods in Solid State Physics

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Contents

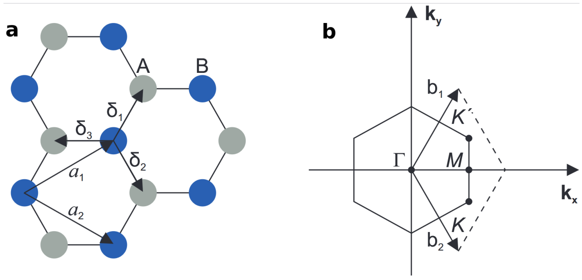

Figure 1: **a** Honeycomb lattice structure of graphene, consisting of two interpenetrating triangular lattices with the lattice unit vectors $\boldsymbol{a_1}$ and $\boldsymbol{a_2}$ and the nearest-neighbor vectors $\boldsymbol{\delta_i}$, $i = 1,2,3$. **b** The corresponding Brillouin zone where the Dirac cones are located at the corner of the first Brillouin zone, the high-symmetry points $\boldsymbol{K}$ and $\boldsymbol{K'}$.

1 Tight binding approximation - the case of graphene

Graphene's lattice is a honeycomb structure composed of two interpenetrating triangular sublattices, denoted as A and B sites. This bipartite lattice is crucial for understanding the electronic behavior of graphene. The unit vectors of the triangular sub-lattices are given by:

$$\boldsymbol{a_{1/2}} = \frac{a}{2}(3, \pm\sqrt{3}). \tag{1}$$

where a is the carbon-carbon distance. The nearest-neighbor vectors are given by:

$$\boldsymbol{\delta_1} = \frac{a}{2}(1, \sqrt{3}) \quad \text{and} \quad \boldsymbol{\delta_2} = \frac{a}{2}(1, -\sqrt{3}) \quad \text{and} \quad \boldsymbol{\delta_3} = -a(1, 0). \tag{2}$$

The two inequivalent high-symmetry points $\boldsymbol{K}$ and $\boldsymbol{K'}$ in $\boldsymbol{k}$ space can, e.g., be described by

$$\boldsymbol{K} = \frac{2\pi}{3\sqrt{3}a}(-\sqrt{3}, -1) \quad \text{and} \quad \boldsymbol{K'} = \frac{2\pi}{3\sqrt{3}a}(+\sqrt{3}, +1). \tag{3}$$

In the simplest form of the tight-binding model, only the interactions between nearest-neighbor atoms are considered. This approximation is often sufficient to capture the essential features of graphene's band structure. The hopping parameter t, which represents the energy associated with an electron hopping from one site to a neighboring site, is typically around 2.7 eV for graphene.

1.1 Diagonalizing the Hamiltonian in k-space

The real space Hamiltonian for graphene in the nearest-neighbor tight-binding model is:

$$H = -t \sum_{\langle ij \rangle} (a_i^\dagger b_j + b_j^\dagger a_i) \tag{4}$$

where $a_i^\dagger$ and $b_j^\dagger$ are the electron creation operators on the A and B sublattices, respectively, and t is the hopping parameter. The summation $\sum_{\langle ij \rangle}$ runs over all pairs $\langle ij \rangle$ of nearest neighbors. The on-site terms are omitted here because they only lead to a rigid shift (identical atoms on each lattice site).

Fourier Transform to Momentum Space: The momentum-space representation is introduced through the Fourier transforms of the creation and annihilation operators. These transforms allow us to move from real space, where the Hamiltonian is expressed in terms of individual lattice sites, to momentum space, where the Hamiltonian is formulated in wavevectors k. In momentum space, the Hamiltonian becomes diagonal, and the electron dynamics are described in terms of the energies associated with momentum eigenstates.
Define the Fourier transforms of the creation and annihilation operators:

$$a_i = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}_i} a_{\boldsymbol{k}} \quad \text{and} \quad b_j = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{R}_j} b_{\boldsymbol{k}}, \tag{5}$$

where N is the number of unit cells, $\boldsymbol{R}_i$ and $\boldsymbol{R}_j$ are the positions of the A and B sites, respectively, and $\boldsymbol{k}$ is the wavevector.
Substitute the Fourier transforms into the real space Hamiltonian:

$$H = -t \sum_{\langle i,j \rangle} \left(\frac{1}{\sqrt{N}} \sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{R}_i} a^\dagger_{\boldsymbol{k}} \frac{1}{\sqrt{N}} \sum_{\boldsymbol{k'}} e^{i\boldsymbol{k'}\cdot\boldsymbol{R}_j} b_{\boldsymbol{k'}} + \text{h.c.} \right). \tag{6}$$

Rewriting the sum over the nearest neighbors: Next, we rewrite the Hamiltonian by reorganizing the sum over nearest neighbors $\langle ij \rangle$: We first sum over all sites i on the A sublattice, and for each site i, we then sum over its three nearest B neighbors, identified by the vectors $\boldsymbol{\delta}$. In other words, the nearest neighbors j are located at positions $\boldsymbol{R}_j = \boldsymbol{R}_i + \boldsymbol{\delta}$, where $\boldsymbol{\delta}$ point from an A-site to the neighboring B-sites. Using that relation, we can then rewrite the Hamiltonian as

$$\begin{aligned} H &= -t \sum_i \sum_{\boldsymbol{\delta}} \left(\frac{1}{N} \sum_{\boldsymbol{k}} \sum_{\boldsymbol{k'}} e^{-i\boldsymbol{k}\cdot\boldsymbol{R}_i} e^{i\boldsymbol{k'}\cdot(\boldsymbol{R}_i+\boldsymbol{\delta})} a^\dagger_{\boldsymbol{k}} b_{\boldsymbol{k'}} + \text{h.c.} \right) && (7) \\ &= -t \sum_i \sum_{\boldsymbol{\delta}} \left(\frac{1}{N} \sum_{\boldsymbol{k}} \sum_{\boldsymbol{k'}} e^{i(\boldsymbol{k'}-\boldsymbol{k})\cdot\boldsymbol{R}_i} e^{i\boldsymbol{k'}\cdot\boldsymbol{\delta}} a^\dagger_{\boldsymbol{k}} b_{\boldsymbol{k'}} + \text{h.c.} \right). && (8) \end{aligned}$$

Orthogonality of Fourier Components: We can simplify the sum by considering the orthogonality of the Fourier components. The orthogonality of the Fourier components implies that:

$$\sum_i e^{i(\boldsymbol{k'}-\boldsymbol{k})\cdot\boldsymbol{R}_i} = N\delta_{\boldsymbol{k},\boldsymbol{k'}}, \tag{9}$$

where $\delta_{\boldsymbol{k},\boldsymbol{k'}}$ is the Kronecker delta, which is 1 if $\boldsymbol{k} = \boldsymbol{k'}$ and 0 otherwise.
Applying the orthogonality condition, we get:

$$\begin{aligned} H &= -t \sum_{\boldsymbol{\delta}} \left(\sum_{\boldsymbol{k}} e^{i\boldsymbol{k}\cdot\boldsymbol{\delta}} a^\dagger_{\boldsymbol{k}} b_{\boldsymbol{k}} + \text{h.c.} \right) && (10) \\ &= -t \sum_{\boldsymbol{k}} \sum_{\boldsymbol{\delta}} \left(e^{i\boldsymbol{k}\cdot\boldsymbol{\delta}} a^\dagger_{\boldsymbol{k}} b_{\boldsymbol{k}} + \text{h.c.} \right) && (11) \end{aligned}$$

Matrix Representation of the Hamiltonian: To represent the tight-binding Hamiltonian of graphene as a matrix, we define the basis vectors as:

$$\psi_{\boldsymbol{k}} = \begin{pmatrix} a_{\boldsymbol{k}} \\ b_{\boldsymbol{k}} \end{pmatrix} \tag{12}$$

The Hamiltonian can then be written in matrix form as:

$$H = \sum_{\boldsymbol{k}} \psi^\dagger_{\boldsymbol{k}} \mathcal{H}(\boldsymbol{k}) \psi_{\boldsymbol{k}}. \tag{13}$$

where $\mathcal{H}(\boldsymbol{k})$ is the 2×2 Hamiltonian matrix:

$$\mathcal{H}(\boldsymbol{k}) = -t \begin{pmatrix} 0 & \Delta(\boldsymbol{k}) \\ \Delta^*(\boldsymbol{k}) & 0 \end{pmatrix}, \tag{14}$$

with $\Delta(\boldsymbol{k}) = \sum_{\boldsymbol{\delta}} e^{i\boldsymbol{k}\cdot\boldsymbol{\delta}}$.

Energy bands: To find the energy eigenvalues, we need to diagonalize the Hamiltonian matrix. The eigenvalues correspond to the energy of the electron states in graphene. The Hamiltonian matrix is a 2×2 matrix, so we solve for the eigenvalues by finding the determinant of $\mathcal{H} - EI$, where E represents the energy eigenvalues and I is the identity matrix. This gives the characteristic equation:

$$\det\left(-t \begin{pmatrix} 0 & \Delta(\boldsymbol{k}) \\ \Delta^*(\boldsymbol{k}) & 0 \end{pmatrix} - E \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\right) = 0 \tag{15}$$

Simplifying:

$$\det\left(\begin{pmatrix} -E & -t\Delta(\boldsymbol{k}) \\ -t\Delta^*(\boldsymbol{k}) & -E \end{pmatrix}\right) = 0 \tag{16}$$

Now, calculate the determinant:

$$E^2 - t^2 \Delta(\boldsymbol{k})\Delta^*(\boldsymbol{k}) = 0 \tag{17}$$

This leads to the energy eigenvalues:

$$E_\pm = \pm t\sqrt{\Delta(\boldsymbol{k})\Delta^*(\boldsymbol{k})}. \tag{18}$$

Inserting the nearest neighbour terms: Next, we insert the nearest neighbour terms in $\Delta(\boldsymbol{k})$: For graphene, the sum over nearest neighbors involves the three vectors from above $\boldsymbol{\delta}_1$, $\boldsymbol{\delta}_2$, and $\boldsymbol{\delta}_3$:

$$\boldsymbol{\delta}_1 = \left(\frac{a}{2}, \frac{\sqrt{3}a}{2}\right), \quad \boldsymbol{\delta}_2 = \left(\frac{a}{2}, -\frac{\sqrt{3}a}{2}\right), \quad \boldsymbol{\delta}_3 = (-a, 0). \tag{19}$$

$$\Delta(\boldsymbol{k}) = e^{i\boldsymbol{k}\cdot\boldsymbol{\delta_1}} + e^{i\boldsymbol{k}\cdot\boldsymbol{\delta_2}} + e^{i\boldsymbol{k}\cdot\boldsymbol{\delta_3}}. \tag{20}$$

Factor out $e^{i\boldsymbol{k}\cdot\boldsymbol{\delta_3}}$:

$$\Delta(\boldsymbol{k}) = e^{i\boldsymbol{k}\cdot\boldsymbol{\delta_3}}\left[1 + e^{i\boldsymbol{k}\cdot(\boldsymbol{\delta_1}-\boldsymbol{\delta_3})} + e^{i\boldsymbol{k}\cdot(\boldsymbol{\delta_2}-\boldsymbol{\delta_3})}\right]. \tag{21}$$

With the above definitions of $\boldsymbol{\delta_i}$ with i = 1,2,3:

$$f(\boldsymbol{k}) = e^{-ik_x a}\left[1 + e^{i\frac{3}{2}k_x a}\left(e^{i\frac{\sqrt{3}}{2}k_y a} + e^{-i\frac{\sqrt{3}}{2}k_y a}\right)\right]. \tag{22}$$

Using the identity $e^{i\theta} + e^{-i\theta} = 2\cos(\theta)$, we get:

$$\Delta(\boldsymbol{k}) = e^{-ik_x a}\left[1 + 2e^{i\frac{3}{2}k_x a}\cos\left(\frac{\sqrt{3}}{2}k_y a\right)\right]. \tag{23}$$

Likewise, $\Delta^*(\boldsymbol{k})$ is defined as:

$$\Delta^*(\boldsymbol{k}) = e^{ik_xa}\left[1 + 2e^{-i\frac{3}{2}k_xa}\cos\left(\frac{\sqrt{3}}{2}k_ya\right)\right]. \tag{24}$$

Now, we compute $\Delta(\boldsymbol{k})\Delta^*(\boldsymbol{k})$:

$$\Delta(\boldsymbol{k})\Delta^*(\boldsymbol{k}) = \left(e^{-ik_xa}\left[1 + 2e^{i\frac{3}{2}k_xa}\cos\left(\frac{\sqrt{3}}{2}k_ya\right)\right]\right)\left(e^{ik_xa}\left[1 + 2e^{-i\frac{3}{2}k_xa}\cos\left(\frac{\sqrt{3}}{2}k_ya\right)\right]\right) \tag{25}$$

$$= \left[1 + 2e^{i\frac{3}{2}k_xa}\cos\left(\frac{\sqrt{3}}{2}k_ya\right)\right]\left[1 + 2e^{-i\frac{3}{2}k_xa}\cos\left(\frac{\sqrt{3}}{2}k_ya\right)\right]. \tag{26}$$

Expanding the product:

$$\Delta(\boldsymbol{k})\Delta^*(\boldsymbol{k}) = 1 + 2e^{i\frac{3}{2}k_xa}\cos\left(\frac{\sqrt{3}}{2}k_ya\right) + 2e^{-i\frac{3}{2}k_xa}\cos\left(\frac{\sqrt{3}}{2}k_ya\right) + 4\cos^2\left(\frac{\sqrt{3}}{2}k_ya\right). \tag{27}$$

Combining the complex conjugate terms:

$$2e^{i\frac{3}{2}k_xa}\cos\left(\frac{\sqrt{3}}{2}k_ya\right) + 2e^{-i\frac{3}{2}k_xa}\cos\left(\frac{\sqrt{3}}{2}k_ya\right) = 4\cos\left(\frac{3}{2}k_xa\right)\cos\left(\frac{\sqrt{3}}{2}k_ya\right). \tag{28}$$

Thus, we have:

$$\Delta(\boldsymbol{k})\Delta^*(\boldsymbol{k}) = 1 + 4\cos\left(\frac{3}{2}k_xa\right)\cos\left(\frac{\sqrt{3}}{2}k_ya\right) + 4\cos^2\left(\frac{\sqrt{3}}{2}k_ya\right). \tag{29}$$

The energy bands are therefore given by:

$$E_\pm(\boldsymbol{k}) = \pm t\sqrt{1 + 4\cos\left(\frac{3}{2}k_xa\right)\cos\left(\frac{\sqrt{3}}{2}k_ya\right) + 4\cos^2\left(\frac{\sqrt{3}}{2}k_ya\right)}. \tag{30}$$

We can also rewrite the term

$$\cos^2\left(\frac{\sqrt{3}}{2}k_ya\right) \tag{31}$$

using the identity $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$. The energy bands can then be expressed as:

$$E_\pm(\boldsymbol{k}) = \pm t\sqrt{3 + f(\boldsymbol{k})}, \tag{32}$$

where

$$f(\boldsymbol{k}) = 2\cos\left(\sqrt{3}k_ya\right) + 4\cos\left(\frac{3}{2}k_xa\right)\cos\left(\frac{\sqrt{3}}{2}k_ya\right). \tag{33}$$

These are two gapless bands that touch at the Dirac points $\boldsymbol{K}$ and $\boldsymbol{K'}$. In other words, the Dirac points are the points in k-space for which $E_\pm(\boldsymbol{k}) = 0$.

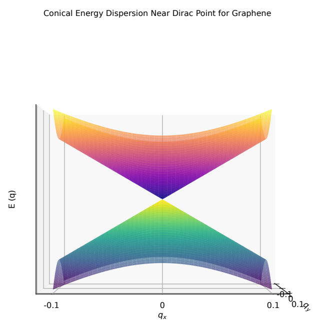

Figure 2: Conical behavior of the bands near the Dirac point. Here, the energy is linear in momentum.

Express in Terms of Pauli Matrices: We can express the Hamiltonian

$$\mathcal{H}(\boldsymbol{k}) = -t \begin{pmatrix} 0 & \Delta(\boldsymbol{k}) \\ \Delta^*(\boldsymbol{k}) & 0 \end{pmatrix} \tag{15}$$

in terms of Pauli matrices by expressing $\Delta(\boldsymbol{k})$ as

$$\Delta(\boldsymbol{k}) = \sum_{\boldsymbol{\delta}} e^{i\boldsymbol{k}\cdot\boldsymbol{\delta}} = \sum_{\boldsymbol{\delta}} \left[\cos(\boldsymbol{k}\cdot\boldsymbol{\delta}) + i\sin(\boldsymbol{k}\cdot\boldsymbol{\delta})\right], \tag{16}$$

so the Hamiltonian reads

$$\mathcal{H}(\boldsymbol{k}) = -t\sum_{\boldsymbol{\delta}} \begin{pmatrix} 0 & \cos(\boldsymbol{k}\cdot\boldsymbol{\delta}) + i\sin(\boldsymbol{k}\cdot\boldsymbol{\delta}) \\ \cos(\boldsymbol{k}\cdot\boldsymbol{\delta}) - i\sin(\boldsymbol{k}\cdot\boldsymbol{\delta}) & 0 \end{pmatrix}. \tag{17}$$

In terms of Pauli matrices, the Hamiltonian is then

$$\mathcal{H}(\boldsymbol{k}) = -t\sum_{\boldsymbol{\delta}} \left[\cos(\boldsymbol{k}\cdot\boldsymbol{\delta})\,\sigma_x - \sin(\boldsymbol{k}\cdot\boldsymbol{\delta})\,\sigma_y\right]. \tag{34}$$

where the Pauli matrices are defined as:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{35}$$

Mass term: We can also add a σ_z term to the Hamiltonian which is then given by:

$$\mathcal{H}_M(\boldsymbol{k}, M) = -t\sum_{\boldsymbol{\delta}} \left[\cos(\boldsymbol{k}\cdot\boldsymbol{\delta})\sigma_x - \sin(\boldsymbol{k}\cdot\boldsymbol{\delta})\sigma_y + M\sigma_z\right] \tag{36}$$

This can also be written in matrix form as:

$$\mathcal{H}_M(\boldsymbol{k}, M) = -t\sum_{\boldsymbol{\delta}} \begin{pmatrix} M & \cos(\boldsymbol{k}\cdot\boldsymbol{\delta}) + i\sin(\boldsymbol{k}\cdot\boldsymbol{\delta}) \\ \cos(\boldsymbol{k}\cdot\boldsymbol{\delta}) - i\sin(\boldsymbol{k}\cdot\boldsymbol{\delta}) & -M \end{pmatrix} \tag{37}$$

From the matrix form of the Hamiltonian, we can see that the additional σ_z term raises the energy of A sites and lowers the energy of B sites, thereby opening a gap in the band structure. We can interpret this as arising from on-site potential terms of the form:

$$\mathcal{H}_{\text{potential}} = M \left(\sum_{\alpha \in \{A \text{ sites}\}} \hat{n}_{a,\alpha} - \sum_{\beta \in \{B \text{ sites}\}} \hat{n}_{b,\beta} \right), \tag{38}$$

where $\hat{n}_{a,\alpha} = \hat{a}^\dagger_\alpha \hat{a}_\alpha$ and $\hat{n}_{b,\beta} = \hat{b}^\dagger_\beta \hat{b}_\beta$. This type of potential can arise when the A and B sublattices consist of different types of atoms, such as in boron nitride. Boron nitride has the same honeycomb lattice structure as graphene, but with boron atoms on A sites and nitrogen atoms on B sites.

1.2 Low energy effective Hamiltonian

Dispersion near K: We now look towards the behavior of $\Delta(\boldsymbol{k})$ near the Dirac point $\boldsymbol{K}$. If we rewrite $\Delta(\boldsymbol{k})$ in terms of $\boldsymbol{k} = \boldsymbol{q} + \boldsymbol{K}$, this means that we express the wave vector $\boldsymbol{k}$ relative to a specific point $\boldsymbol{K}$ in reciprocal space. This is particularly useful near the Dirac points in graphene, as the energy exhibits a linear dispersion relation there.

$$\Delta(\boldsymbol{K} + \boldsymbol{q}) = e^{-iK_x a} e^{-iq_x a} \left[1 + 2e^{i\frac{3}{2}(K_x + q_x)a} \cos\left(\frac{\sqrt{3}}{2}(K_y + q_y)a \right) \right]. \tag{39}$$

At the Dirac point $\boldsymbol{K} = (K_x, K_y)$ we have $K_x = -\frac{2\pi}{3a}$ and $K_y = -\frac{2\pi}{3\sqrt{3}a}$.

$$\Delta(\boldsymbol{K} + \boldsymbol{q}) = e^{-iK_x a} e^{-iq_x a} \left[1 - 2e^{i\frac{3}{2}q_x a} \cos\left(-\frac{\pi}{3} + \frac{\sqrt{3}}{2} q_y a \right) \right] \tag{40}$$

We now expand $\Delta(\boldsymbol{K} + \boldsymbol{q})$ to first order in $\boldsymbol{q}$ about $\boldsymbol{q} = 0$. For small q_x, we expand $e^{-iq_x a}$ as:

$$e^{-iq_x a} \approx 1 - iq_x a + O(q_x^2) \tag{41}$$

Similarly, expand $e^{i\frac{3}{2}q_x a}$ to first order in q_x:

$$e^{i\frac{3}{2}q_x a} \approx 1 + i\frac{3}{2} q_x a + O(q_x^2) \tag{42}$$

For small q_y, we expand $\cos\left(-\frac{\pi}{3} + \frac{\sqrt{3}}{2} q_y a\right)$ as:

$$\cos\left(-\frac{\pi}{3} + \frac{\sqrt{3}}{2} q_y a \right) \approx \cos\left(-\frac{\pi}{3}\right) + \frac{\sqrt{3}}{2} q_y a \sin\left(\frac{\pi}{3}\right) \tag{43}$$

Since $\cos\left(-\frac{\pi}{3}\right) = \frac{1}{2}$ and $\sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$, we get:

$$\cos\left(-\frac{\pi}{3} + \frac{\sqrt{3}}{2} q_y a \right) \approx \frac{1}{2} + \frac{3}{4} q_y a \tag{44}$$

Substitute the expansions into the original expression for $\Delta(\boldsymbol{K} + \boldsymbol{q})$:

$$\Delta(\boldsymbol{K} + \boldsymbol{q}) \approx e^{-iK_x a} (1 - iq_x a) \left[1 - 2\left(1 + i\frac{3}{2} q_x a\right) \left(\frac{1}{2} + \frac{3}{4} q_y a\right) \right] \tag{45}$$

Now expand the terms inside the brackets:

$$\left(1 + i\frac{3}{2}q_x a\right)\left(\frac{1}{2} + \frac{3}{4}q_y a\right) = \frac{1}{2} + \frac{3}{4}q_y a + i\frac{3}{4}q_x a + i\frac{9}{8}q_x q_y a^2 \tag{46}$$

Multiplying by -2 and adding 1:

$$1 - 2\left(\frac{1}{2} + \frac{3}{4}q_y a + i\frac{3}{4}q_x a + i\frac{9}{8}q_x q_y a^2\right) = -\frac{3}{2}q_y a - i\frac{3}{2}q_x a + i\frac{9}{4}q_x q_y a^2 \tag{47}$$

Now substitute this back into the expression for $\Delta(\boldsymbol{K}+\boldsymbol{q})$ and only keep terms linear in q_x and q_y

$$\Delta(\boldsymbol{K}+\boldsymbol{q}) \approx e^{-iK_x a}\left[-\frac{3}{2}q_y a - i\frac{3}{2}q_x a + O(q_x^2, q_y^2, q_x q_y)\right] \tag{48}$$

To first order, we are thus left with:

$$\Delta(\boldsymbol{K}+\boldsymbol{q}) \approx -e^{-iK_x a}\left[\frac{3}{2}a\left(q_y + iq_x\right)\right] \tag{49}$$

$$= -ie^{-iK_x a}\frac{3a}{2}(q_x - iq_y) \tag{50}$$

The phase of $\Delta(\boldsymbol{K}+\boldsymbol{q})$ carries no physical significance (since, for example, the energy bands are given by $E_\pm(\boldsymbol{K}+\boldsymbol{q}) = \pm t\sqrt{\Delta(\boldsymbol{K}+\boldsymbol{q})\Delta^*(\boldsymbol{K}+\boldsymbol{q})}$), so it is convenient to ignore the phase of $ie^{-iK_x a}$. We thus have

$$\Delta(\boldsymbol{K}+\boldsymbol{q}) = -\frac{3a}{2}(q_x - iq_y). \tag{51}$$

About the Dirac point $\boldsymbol{K}$, the Hamiltonian is thus

$$\mathcal{H}(\boldsymbol{K}+\boldsymbol{q}) = \hbar v_F \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix}, \tag{52}$$

where

$$v_F = \frac{3at}{2\hbar} \tag{53}$$

is the Fermi velocity. We can express this in terms of Pauli matrices as

$$\mathcal{H}(\boldsymbol{K}+\boldsymbol{q}) = \hbar v_F(q_x\sigma_x + q_y\sigma_y) = \hbar v_F \boldsymbol{q}\cdot\boldsymbol{\sigma}. \tag{54}$$

Dispersion near K': Similarly, we can define the relative momentum $\boldsymbol{q} = \boldsymbol{k} - \boldsymbol{K'}$ and expand around the Dirac point $\boldsymbol{K'}$:

$$\Delta(\boldsymbol{K'}+\boldsymbol{q}) = -\frac{3a}{2}(q_x + iq_y). \tag{55}$$

So the Hamiltonian about this Dirac point is

$$\mathcal{H}(\boldsymbol{K'}+\boldsymbol{q}) = \hbar v_F \begin{pmatrix} 0 & q_x + iq_y \\ q_x - iq_y & 0 \end{pmatrix} = \hbar v_F(q_x\sigma_x - q_y\sigma_y) = \hbar v_F \bar{\boldsymbol{q}}\cdot\boldsymbol{\sigma}. \tag{56}$$

with the vector

$$\bar{\boldsymbol{q}} \equiv \begin{pmatrix} q_x \\ -q_y. \end{pmatrix}, \tag{57}$$

Linear dispersion relation: From the Hamiltonian near the Dirac points, we find that the energy bands near the Dirac points are given by

$$E_{\pm}(\boldsymbol{q}) = \hbar v_F |\boldsymbol{q}|. \tag{58}$$

Near a Dirac point, the dispersion relation is therefore linear in momentum, so the energy bands form a cone (called a Dirac cone) with the vertex lying on the Dirac point.

Mass term: If we add the σ_z term that we introduced earlier, the Hamiltonian near the $\boldsymbol{K}$ Dirac points will, e.g., be of the form

$$\mathcal{H}_M(\boldsymbol{q}) = \hbar v_F (q_x \sigma_x + q_y \sigma_y + M\sigma_z) = \hbar v_F \begin{pmatrix} M & q_x - iq_y \\ q_x + iq_y & -M \end{pmatrix}. \tag{59}$$

As we saw, this additional term gaps the energies, so that near the Dirac points, the energies become

$$E_{M,\pm} = \pm \hbar v_F \sqrt{q_x^2 + q_y^2 + M^2}. \tag{60}$$

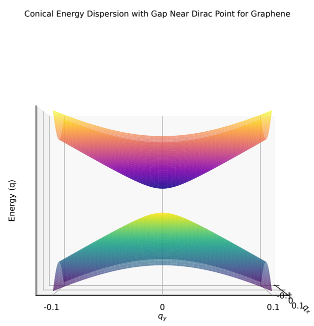

Figure 3: Bands near the Dirac point when including a σ_z term, which creates a gap.

1.3 Eigenstates and chirality in Graphene

In the previous paragraphs, we have shown that the effective massless Dirac Hamiltonian in graphene is given by:

$$\mathcal{H} = \hbar v_F (\sigma_x q_x + \tau_z \sigma_y q_y), \tag{61}$$

where:

- v_F is the Fermi velocity.
- $\tau_z = \pm 1$ denotes the valley index ($\boldsymbol{K}$ or $\boldsymbol{K'}$).
- σ_x and σ_y are the Pauli matrices.

- q_x and q_y are the wavevector components relative to the Dirac points.

To find the eigenstates, we solve the eigenvalue equation:

$$\mathcal{H}\boldsymbol{\psi} = E\boldsymbol{\psi} \tag{62}$$

For the Hamiltonian at the $\boldsymbol{K}$ point ($\tau_z = 1$):

$$\mathcal{H} = \hbar v_F(\sigma_x q_x + \sigma_y q_y) \tag{63}$$

To find the eigenvalues, we solve the characteristic equation:

$$\det(\mathcal{H} - EI) = 0 \tag{64}$$

$$\Rightarrow \det\begin{pmatrix} -E & \hbar v_F(q_x - iq_y) \\ \hbar v_F(q_x + iq_y) & -E \end{pmatrix} = E^2 - (\hbar v_F)^2(q_x^2 + q_y^2) = 0 \tag{65}$$

$$\Rightarrow E = \pm\hbar v_F\sqrt{q_x^2 + q_y^2} = \pm\hbar v_F|\boldsymbol{q}| \tag{66}$$

The eigenstates $\boldsymbol{\psi}$ are two-component spinors:

$$\boldsymbol{\psi} = \begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} \tag{67}$$

Substituting these into the Hamiltonian, we get:

$$\hbar v_F\begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix}\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} = E\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} = \pm\hbar v_F|\boldsymbol{q}|\begin{pmatrix} \psi_A \\ \psi_B \end{pmatrix} \tag{68}$$

Case 1: $E = +\hbar v_F|\boldsymbol{q}|$

$$\begin{cases} (q_x - iq_y)\psi_B = |\boldsymbol{q}|\psi_A \\ (q_x + iq_y)\psi_A = |\boldsymbol{q}|\psi_B \end{cases} \Rightarrow \psi_B = \frac{q_x + iq_y}{|\boldsymbol{q}|} \tag{69}$$

Let $\theta = \arg(q_x + iq_y)$, so $\frac{q_x+iq_y}{|\boldsymbol{q}|} = e^{i\theta}$. Choosing $\psi_A = e^{-i\frac{\theta}{2}}$ the eigenstate reads

$$\boldsymbol{\psi}^{(+)} = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-i\frac{\theta}{2}} \\ e^{+i\frac{\theta}{2}} \end{pmatrix} \tag{70}$$

Case 2: $E = -\hbar v_F|\boldsymbol{q}|$

$$\begin{cases} (q_x - iq_y)\psi_B = -|\boldsymbol{q}|\psi_A \\ (q_x + iq_y)\psi_A = -|\boldsymbol{q}|\psi_B \end{cases} \Rightarrow \psi_B = -\frac{(q_x + iq_y)}{|\boldsymbol{q}|}\psi_A \tag{71}$$

So:

$$\boldsymbol{\psi}^{(-)} = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-i\frac{\theta}{2}} \\ -e^{i\frac{\theta}{2}} \end{pmatrix} \tag{72}$$

The eigenstates are

$$\psi_{\boldsymbol{K}}^{(\pm)} = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{-i\frac{\theta}{2}} \\ \pm e^{i\frac{\theta}{2}} \end{pmatrix}, \quad \theta = \arg(q_x + iq_y) \tag{73}$$

For the Hamiltonian at the $\boldsymbol{K'}$ point ($\tau_z = -1$) we can similarly calculate the eigenvalues and eigenstates. The eigenenergies are the same as for $\boldsymbol{K}$

$$\Rightarrow E = \pm\hbar v_F \sqrt{q_x^2 + q_y^2} = \pm\hbar v_F |\boldsymbol{q}| \tag{74}$$

The eigenstates read

$$\psi_{\boldsymbol{K'}}^{(\pm)} = \frac{1}{\sqrt{2}} \begin{pmatrix} e^{i\frac{\theta}{2}} \\ \pm e^{-i\frac{\theta}{2}} \end{pmatrix}, \quad \theta = \arg(q_x + iq_y). \tag{75}$$

Chirality near the Dirac points Now that we have defined the Hamiltonian near the Dirac points, we consider the chirality associated with the $\boldsymbol{K}$ and $\boldsymbol{K'}$ points next. The sign difference in the terms involving q_x and q_y in the Hamiltonian determines the chirality of the system. At the $\boldsymbol{K}$ point, the Hamiltonian is:

$$\mathcal{H}(\boldsymbol{K} + \boldsymbol{q}) = \hbar v_F \begin{pmatrix} 0 & q_x - iq_y \\ q_x + iq_y & 0 \end{pmatrix} = \hbar v_F (q_x \sigma_x + q_y \sigma_y) = \hbar v_F \boldsymbol{q} \cdot \boldsymbol{\sigma}. \tag{76}$$

The sign of the terms in the Hamiltonian indicates that the momentum space near the $\boldsymbol{K}$ point is right-handed, or positive chirality. This means that an electron at the $\boldsymbol{K}$ point moves in a clockwise direction in momentum space. At the $\boldsymbol{K'}$ point, the Hamiltonian is:

$$\mathcal{H}(\boldsymbol{K'} + \boldsymbol{q}) = \hbar v_F \begin{pmatrix} 0 & q_x + iq_y \\ q_x - iq_y & 0 \end{pmatrix} = \hbar v_F (q_x \sigma_x - q_y \sigma_y) = \hbar v_F \bar{\boldsymbol{q}} \cdot \boldsymbol{\sigma}. \tag{77}$$

For the $\boldsymbol{K'}$ point, the opposite sign appears in the Hamiltonian, which corresponds to left-handed, or negative chirality. This indicates that an electron at the $\boldsymbol{K'}$ point moves in a counter-clockwise direction in the momentum space. Thus, the difference in the sign of the terms in the Hamiltonian leads to the distinction between the two valleys in momentum space. The two valleys, $\boldsymbol{K}$ and $\boldsymbol{K'}$, are associated with opposite chirality, which can have important consequences for electronic properties, such as the direction of current flow in response to an applied magnetic field (the so-called valley Hall effect).